



Why Ethics?

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Ethics background required: None. This module motivates the need for ethics education in computer science and/or data science.

Subject matter referred to in this module: What do computer scientists/data scientists do?

Placement in overall ethics curriculum:

Academic year: First module in the first year of curriculum use. Ideally, this is the first semester of the freshman year. We would recommend that the students are 3-4 weeks into the first semester when this module is administered.

Recommended previous modules: None

Recommended follow-up modules: Virtues module, but any module can follow

Time required:

- In class: 25 minutes

Learning objectives:

- Students gain a better understanding of how computer scientists > and/or data scientists contribute to the world.
- Students have the opportunity to think about the responsibility that > rests on their shoulders should they pursue a career in computer > and/or data science.
- Students see the need for ethics education in computer and/or data > science.
- Students understand what the study of ethics is

Module overview:

Ethical issues to be considered:

This module briefly considers many ethical issues in computing and data science. The actual issues covered will depend on the professor and class contributions.

Exercise flow:



Students are introduced to the variety of contributions that > Computer and Data Scientists make to the world

- Student groups consider what could go wrong in the areas of Data and > Computer Science that they just learned about
- Ethics as a field of study is defined
- Explanation that future modules will help provide tools for and give > students practice in recognizing and making decisions about > potential ethical issues that they could encounter in their future > work.

Preparation: Read through the entire module. Make copies of the job descriptions for computer and data scientists. Alternatively, make the descriptions available online.

Instructors Guide

Lesson plan

Introduction (10 min)

Hand out (or point students to) the list below of jobs that computer and data scientists might be hired to do. Give them a moment to look it over, and ask if they would like to add anything.

Computer Science

- Software developer
 - Games
 - Social media
 - Airplane/Car navigational systems
 - Predictive algorithms (who is a good candidate for a loan or > job, predicting future weather or erosion patterns, predicting > if a person who is arrested is a flight risk)
 - Science applications (finding genetically-related diseases, > analyzing tumor progression/reduction, drug discovery)
 - Compilers/operating systems
- Database Administrator
 - Design systems to store and retrieve information
- Computer hardware engineer
 - Design the next phone, computer or tablet
 - Create/test circuit boards, routers, memory devices



integrated ethics LABS

- Create ways for disabled to move and/or communicate
- Information security analyst
 - Detect cyber attacks
 - Prevent cyber attacks
- Web Developer
 - Create the technical structure for websites
 - Make sure that web pages are accessible and easily downloadable
 - Structure sites to maximize the number of page views and > visitors through search engine optimization.

Data Science

- Data Analytics
 - Help develop strategies to make better business decisions
 - Create/use programs to help individuals invest money
 - Track and analyze website usage patterns
- Data architect
 - Find, clean and organize data
 - Look for patterns in data collected
- Machine learning
 - Develop predictive algorithms based on what has happened in the > past (medicine, business, job/student success, economy, > climate change, disease spread etc.)
- Statistician
 - Interpret, analyze, and report statistical information (could be > related to research in any field: business, psychology, > sociology, medicine..)

Activity (15 min)

Read or summarize to students: It should be apparent from the list that computer and data scientists can hold jobs that are very influential in society. With such power, people in these positions can bring great good, or, if they are not careful, great harm. In the next 10 minutes, brainstorm with your group things that could go wrong in any of these areas if they are not handled carefully. Each group should choose a note-taker and spokesperson to that your ideas can be shared with the larger group later.

Reflection as a class (10 min)



After 10 minutes, return the students to the larger group to share their thoughts.

Here are some ideas if the students have not been about to come up with many:

- Games can be violent and/or addictive
- Social media can also be addictive and also divisive. It can > propagate wrong ideas and bullying, can enable predators, and may > trivialize or commoditize relationships.
- Security specialists could use their training for wrong purposes.
- Hardware engineers could produce too much e-waste and/or use too > much energy.
- Statisticians could skew results to force desired outcomes.
- Data architects could clean data in a way that eliminates diversity > from the data.
- Web designers can embed software to collect unauthorized data.

Wrap up

According to author Sara Baase, in her book *A Gift of Fire*, “Ethics is the study of what it means to do the right thing.” While this is a simple definition, it turns out that the study of ethics can be a complex topic. However, it is certainly one that computer and data scientists, because of the nature of their work, need to carefully consider as they are trained and then move forward into their careers.

If your university has a plan to use future modules, mention that students will have opportunities in other classes to learn tools to help them as computer/data scientists make ethical decisions on implementations and to anticipate detrimental outcomes.

Assessment (to be included later on a quiz, paper, or exam to determine if LOs were reached)

1. Name 3 jobs that you as a computer or data scientist could hold in > the future
2. What is ethics?
3. Give an example of why it is important for computer and data > scientists to study ethics.

Note** This lesson is easily adapted to an asynchronous online environment where a student would complete the assignment individually and then post some of their ideas while also responding to the ideas of others.

Student handouts for group discussions are attached

Potential Jobs for Computer and Data Scientists

Computer Science

- Software developer



- Airplane/Car navigational systems
- Predictive algorithms (who is a good candidate for a loan or > job, predicting future weather or erosion patterns, predicting > if a person who is arrested is a flight risk)
- Science applications (finding genetically-related diseases, > analyzing tumor progression/reduction, drug discovery)
- Compilers/operating systems
- Database Administrator
 - Design systems to store and retrieve important information
- Computer hardware engineer
 - Design the next phone, computer or tablet
 - Create/test circuit boards, routers, memory devices
 - Create ways for disabled to move and/or communicate
- Information security analyst
 - Detect cyber attacks
 - Prevent cyber attacks
- Web Developer
 - Create the technical structure for websites
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Data Science

- Data Analytics
 - Help develop strategies to make better business decisions
 - Create/use programs to help individuals invest money
 - Track and analyze website usage patterns
- Data architect
 - Find, clean and organize data
 - Look for patterns in data collected



- Machine Learning

- Develop predictive algorithms based on what has happened in the > past (medicine, business, job/student success, economy, > climate change, disease spread, loan repayment, criminal > detainee flight risk etc.)

- Statistician

- Interpret, analyze, and report statistical information (could be > related to research in any field: business, psychology, > sociology, medicine..)

Others?

Group work:

Discuss with your group things that could go wrong as these jobs are performed, or the products developed. Two examples to get you started: an information security analyst could use their skills to hack rather than protect, and a data analytics person could embed software to collect unauthorized data.

Make sure to pick a group note-taker and spokesperson so that you will be prepared to share your ideas with the class.

Bibliography